Mostafa Elsayed Masoud
Corresponding author
moustafa.masoud@gmail.com
31 July 2026

To the Editorial Office
Algorithms (MDPI)

Manuscript: *DT-GSK: Dimension-Tiered Adaptive Control and Deterministic Refinement for Gaining-Sharing Knowledge Optimization*
Authors: Mostafa Elsayed Masoud, Heba Sayed Mohamed Roshdy, Ali Wagdy Mohamed
Corresponding author: Mostafa Elsayed Masoud (moustafa.masoud@gmail.com)

Dear Editors,

We are pleased to submit our manuscript, *DT-GSK: Dimension-Tiered Adaptive Control and Deterministic Refinement for Gaining-Sharing Knowledge Optimization*, for consideration in *Algorithms*.

To our knowledge, DT-GSK attains the best overall CEC2017 Friedman mean rank on the seven-algorithm GSK-family panel (2.48, the unweighted mean of the four per-dimension ranks — a descriptive aggregate; eGSK is second at 2.96; the two are never Nemenyi-separable at any CEC2017 dimension, DT-GSK is second behind eGSK at $D = 30$ and on CEC2011; the CEC2011 loss is Holm-significant, and CEC2017 was configuration-selection exposed), evaluated under a release-locked protocol, with byte-stable determinism for DT-GSK in the declared supported environment. The evaluation spans five suites: alongside CEC2017, CEC2011 and the CEC2013 second comparison suite, the study reports CEC2020 under a statistical analysis plan committed to the repository before any CEC2020 result existed, and a family-internal CEC2013LSGO panel at $D = 1000$. We report the unfavorable outcomes with the favorable ones: on CEC2020 — the competition in which our comparator AGSK was the runner-up — DT-GSK places fourth of seven, consistent with the boundary condition the registered directional expectation predicted for a method whose dimension-gated subsystems are inactive at $D \leq 20$; on CEC2013LSGO, DT-GSK and AGSK share the best descriptive rank and paired tests do not separate them. The paper makes three contributions. First, a **deterministic, RNG-free eigenframe final polish** executed once in the final budget slice on a learned interaction eigenbasis; no convergence guarantee is claimed. Second, a **dimension-tiered adaptive scaffold** — a bandit-style operator-configuration selector with acceptance-gated arm pruning, a tier-floored population-size reduction schedule, a hard-capped stagnation escape, and a deep-stall restart that preserves the global best — presented as an honestly labeled modified/original composite over published GSK-family mechanisms, never as a new base operator. Third, the **evaluation-integrity infrastructure** that makes the study checkable: a 13-substream, prefix-locked random-number layer, a paired optimizer-independent seed schedule, a hash-frozen configuration, and promoted, read-only evidence releases to which every reported number is bound. A supporting mechanism — the **interaction-structure memory**, a decaying, confidence- and evidence-gated coordinate-pair graph accumulated solely from strictly improving accepted moves at no extra objective evaluations — supplies the polish eigenbasis and the linkage blocks of the block crossover at dimensions of 50 and above; its direct isolation is reported transparently as a controlled negative result (no significant standalone benefit at its active tiers), and we do not claim it recovers the objective's separability structure. All comparative claims in the manuscript are scoped to this GSK-family panel; we make no field-wide performance claims.

We believe this fits *Algorithms* well: the manuscript is an algorithmic contribution with a fully specified mechanism, a reproducible artifact chain, and panel-scoped empirical evidence. In the same spirit, the paper includes a direct component isolation of the interaction-structure memory and reports its outcome transparently: the isolation finds no detectable standalone benefit from the memory at its active tiers, and we are careful not to over-attribute performance to any mechanism the isolation does not separate — the deterministic compass endgame is evaluated as a whole, and the added value of the learned basis over coordinate or random directions remains unresolved.

We confirm that this manuscript is original, has not been published previously, and is not under consideration by any other journal. All authors have read and approved the submitted version. For transparency, we note that A.W.M. is the originator of the baseline GSK algorithm and a co-author of several of the family variants used as comparators, one of which (eGSK) is also co-authored by H.S.M.R.; these relationships are declared in the manuscript's conflicts-of-interest statement. During the preparation of this manuscript, the authors used a large language model (Claude Opus 4.6, 4.8 and 5.0, Anthropic) to assist with language editing, rephrasing, the drafting of descriptive and expository text, and software-engineering support during implementation and tooling work, in accordance with the MDPI policy on the use of generative artificial intelligence; all scientific content was produced by the authors' own deterministic analysis pipeline, and the authors have reviewed, verified, and edited all AI-assisted text and take full responsibility for the content of this publication.

Thank you for your consideration. We look forward to the reviewers' feedback.

Sincerely,

Mostafa Elsayed Masoud (corresponding author)
Heba Sayed Mohamed Roshdy
Ali Wagdy Mohamed